

CELL BIOLOGY - COMPREHENSIVE ADVANCED NOTES

Every slide covered • Slides remain the backbone • Textbook coverage substantially expanded • No external sources

NOTESGPT FORMAT FOLLOWED

Advanced bullet-point notes focused on exam-relevant material.

Key concepts and vocabulary are bolded.

Every explanatory bullet is followed by a separate WHY in simple language.

Medical terminology is followed by a bracketed plain-language explanation when first introduced.

Each section ends with a summary table.

Slide diagrams and image-only slides are explicitly incorporated or identified as visual reinforcement.

Textbook supplementation is clearly labeled, with a dedicated expansion portion covering major Chapter 2 material that the slides omit or treat only briefly.

Source files

Cell 1 - updated May 26 (27 slides) | Cell 2 - updated May 26 (42 slides) | Chapters 2 & 3 PDF (provided textbook material)

1. Cell Organization & Specialized Cellular Functions

Primary slides: Cell 1, Slides 1-3. Textbook supplement: Chapter 2, Cell Differentiation + Table 2-1.

Slide-derived core

The cell is composed of two basic parts: **cytoplasm** [the cell material outside the nucleus] and the **nucleus** [the DNA-containing control compartment]. *Cell 1, Slide 3*

WHY: This is the most basic map of a eukaryotic cell: one compartment stores and manages genetic information, while the rest of the cell performs most day-to-day work.

The **cytoplasm** contains a fluid component called **cytosol** [cell fluid], and the cytosol contains or bathes organelles. *Cell 1, Slide 3*

WHY: Think of cytosol as the liquid workspace surrounding specialized cellular “machines.”

Movement is a major function of muscle and other contractile cells. *Cell 1, Slide 2*

WHY: These cells devote much of their internal machinery to generating force.

Formation of **adhesive and tight junctions** is emphasized in epithelial cells. *Cell 1, Slide 2*

WHY: Epithelial sheets must stay connected so they can form continuous barriers.

Synthesis and secretion of **extracellular matrix (ECM)** [material outside cells that supports tissues] is emphasized in fibroblasts and cells of bone and cartilage. *Cell 1, Slide 2*

WHY: These cells build the surrounding scaffold of connective tissues.

Conversion of physical and chemical stimuli into **action potentials** is emphasized in neurons and sensory cells. *Cell 1, Slide 2*

WHY: These cells are specialized to turn a stimulus into an electrical signal.

Synthesis and secretion of enzymes is emphasized in cells of digestive glands. *Cell 1, Slide 2*

WHY: Digestive glands must manufacture large quantities of proteins that chemically break down food.

Synthesis and secretion of mucous substances is emphasized in mucous-gland cells. *Cell 1, Slide 2*

WHY: Mucus-producing cells are organized around making and exporting protective secretions.

Synthesis and secretion of **steroids** [lipid-derived hormones] is emphasized in some adrenal, testis, and ovary cells. *Cell 1, Slide 2*

WHY: Steroid-producing cells need abundant lipid-processing machinery.

Ion transport is emphasized in cells of kidney and salivary-gland ducts. *Cell 1, Slide 2*

WHY: These cells regulate fluid composition by moving charged particles across membranes.

Intracellular digestion is emphasized in macrophages [large phagocytic immune cells] and some white blood cells. *Cell 1, Slide 2*

WHY: These cells swallow material and therefore need strong internal digestive machinery.

Lipid storage is emphasized in fat cells. *Cell 1, Slide 2*

WHY: Fat cells specialize in storing energy-rich lipid droplets.

Metabolite absorption is emphasized in cells lining the intestine. *Cell 1, Slide 2*

WHY: Intestinal cells are built to move digested nutrients from the lumen into the body.

Textbook supplement

The textbook describes **differentiation** [cell specialization] as a process in which cells express selected gene sets, change their cytoplasmic activities, and often change shape to become efficient at specialized functions. *Textbook, Ch. 2*

WHY: Cells contain the same general cellular toolkit, but differentiation changes which parts of that toolkit are emphasized.

The textbook stresses that many general cellular functions can be performed by most cells, but specialized cells greatly expand their capacity for one or more of those functions. *Textbook, Ch. 2*

WHY: Specialization is usually a matter of degree: every cell may perform a task a little, while a specialized cell performs it extremely well.

Section Summary Table

Concept	Must know	Exam connection
Basic cell plan	Cytoplasm + nucleus	Know where organelles vs DNA-related processes are located.
Differentiation	Selective gene expression produces specialized structure/function.	Cell shape and organelle abundance often reflect workload.
Secretory cells	Digestive/mucous/steroid-producing cells emphasize secretion.	Match the product with the organelles used to make it.
Transport/absorption cells	Kidney/salivary ducts move ions; intestine absorbs metabolites.	Membrane proteins and surface area are high-yield.
Phagocytic cells	Macrophages and some WBCs perform intracellular digestion.	Endocytosis + lysosomes become important later.

2. Plasma Membrane Architecture, Proteins & Glycocalyx

Primary slides: Cell 1, Slides 4-8 and 26. Textbook supplement: Chapter 2, The Plasma Membrane.

Slide-derived core

The **plasma membrane** / **plasmalemma** [the cell's outer boundary] is the outermost component of the cell. *Cell 1, Slide 4*

WHY: It separates the cell's internal environment from the extracellular environment.

All eukaryotic cells are enclosed by a membrane containing **phospholipids**, **cholesterol**, **proteins**, and **oligosaccharide chains** attached to phospholipids and proteins. *Cell 1, Slide 4*

WHY: A membrane needs lipids for the barrier, proteins for work, and carbohydrates for recognition and interaction.

Membranes are approximately **7.5-10 nm** thick and are visible only with the electron microscope. *Cell 1, Slide 4*

WHY: This is far below the resolving power of routine light microscopy.

The plasma-membrane diagram emphasizes four functions: **communication**, **intercellular connection**, **physical barrier**, and **selective permeability**. *Cell 1, Slide 5*

WHY: The membrane is not only a wall; it is also a gate, communication surface, and structural connection point.

Cholesterol modulates the fluidity and movement of membrane components, and the lipid composition of the two halves of the bilayer is different. *Cell 1, Slide 6*

WHY: The membrane must remain flexible but organized; cholesterol helps control that balance, and asymmetry lets each surface perform different jobs.

Carbohydrates on **glycoproteins** and **glycolipids** act as receptors and participate in cell adhesion, cell recognition, and response to protein hormones. *Cell 1, Slide 6*

WHY: Sugar chains on the outside of the cell function like labels and docking sites.

Membrane proteins are divided into **integral proteins** [embedded in the bilayer] and **peripheral proteins** [more loosely associated with one membrane surface]. **Multipass transmembrane proteins** cross the bilayer repeatedly. *Cell 1, Slide 7*

WHY: Proteins that must move substances or signals across the membrane need to be embedded in it; others can work from one surface.

The **glycocalyx** [carbohydrate-rich outer cell coat] consists of carbohydrate chains linked to membrane proteins and lipids and participates in cell recognition and attachment to other cells and extracellular molecules. *Cell 1, Slide 8*

WHY: The glycocalyx gives the cell an externally facing identity and attachment layer.

Textbook supplement

Membrane phospholipids are **amphipathic** [one water-loving part and one water-fearing part]: polar hydrophilic heads face water, while nonpolar hydrophobic fatty-acid tails form the membrane interior. *Textbook, Ch. 2*

WHY: This arrangement spontaneously creates a stable bilayer with a water-repelling center.

The textbook describes the **fluid mosaic model**: proteins form a mobile mosaic within a fluid lipid bilayer, although cytoskeletal attachments and junctions can restrict protein movement. *Textbook, Ch. 2*

WHY: A dynamic membrane can reorganize transporters and receptors when the cell needs them.

The textbook adds that **integrins** link plasma-membrane proteins to both the cytoskeleton and extracellular-matrix components. *Textbook, Ch. 2*

WHY: Integrins mechanically and functionally connect the inside of the cell with its surroundings.

The textbook describes **lipid rafts** [membrane patches enriched in cholesterol and saturated fatty acids] as regions that help keep selected signaling proteins near one another. *Textbook, Ch. 2*

WHY: Grouping related proteins in the same patch can make signaling reactions more efficient.

EXAM CHECK - In TEM, the membrane appears trilaminar. What is the central poorly stained region primarily

responsible for?

Correct answer: Creation of a barrier to water-soluble molecules.

WHY: The poorly stained middle corresponds to the hydrophobic lipid interior, which repels water-soluble substances.

Cell 1, Slide 26

Section Summary Table

Concept	Must know	Exam connection
Phospholipid bilayer	Hydrophilic surfaces + hydrophobic center	Hydrophobic center is the key barrier to water-soluble molecules.
Cholesterol	Controls membrane fluidity/mobility	Think stability without rigidity.
Integral vs peripheral	Embedded vs surface-associated proteins	Transmembrane channels/receptors are integral.
Glycocalyx	External carbohydrate coat	Recognition, adhesion, receptor interactions.
Fluid mosaic	Lipids/proteins are dynamic	Protein movement can be constrained by cytoskeleton/junctions.

3. Membrane Transport, Endocytosis, LDL Uptake & Exocytosis

Primary slides: Cell 1, Slides 9-15. Textbook supplement: Chapter 2, Transmembrane Transport + Transport by Vesicles.

Transport across the membrane

Ions such as Na^+ , K^+ , and Ca^{2+} cross the cell membrane through integral membrane proteins. *Cell 1, Slide 9*

WHY: Charged ions cannot freely pass through the hydrophobic membrane core, so they need protein pathways.

Passive transport includes simple diffusion, facilitated diffusion, and osmosis. *Cell 1, Slide 9*

WHY: Passive processes use an existing concentration difference rather than direct ATP expenditure.

Active transport uses ion pumps and ATP; the slide lists primary active transport, secondary active transport, symport, and antiport. *Cell 1, Slide 9*

WHY: Active transport lets a cell maintain internal concentrations that differ from the environment.

Endocytosis

Endocytosis [bulk uptake into the cell] occurs when the membrane folds and fuses to form vesicles that enclose transported material. *Cell 1, Slide 10*

WHY: Large particles and bulk fluid cannot pass directly through the lipid bilayer, so the membrane packages them into vesicles.

The three general types of endocytosis listed are **phagocytosis**, **fluid-phase endocytosis (pinocytosis)**, and **receptor-mediated endocytosis**. *Cell 1, Slide 10*

WHY: The cell can engulf large particles, sample fluid, or selectively capture specific ligands.

Phagocytosis ["cell eating"] is ingestion of particles such as bacteria or dead-cell remnants; the resulting vesicle is a **phagosome** [a membrane bag containing the engulfed particle]. *Cell 1, Slide 11*

WHY: The phagosome isolates a large target so it can later be digested safely.

Pinocytosis / fluid-phase endocytosis ["cell drinking"] forms small membrane invaginations that trap extracellular fluid; slide vesicles are about **80 nm** in diameter. *Cell 1, Slide 12*

WHY: This lets the cell take in a sample of the surrounding fluid and dissolved material.

In **receptor-mediated endocytosis**, ligand binding causes receptors to cluster in **coated pits**; the major coat polypeptide shown is **clathrin**. A coated vesicle carries the ligand and receptor into the cell. *Cell 1, Slide 13*

WHY: Receptors make uptake selective and clathrin helps bend the membrane into a vesicle.

The LDL diagram shows **LDL** binding LDL receptors in a clathrin-coated pit, coated-vesicle formation, entry into an endosome, receptor recycling back to the surface, and delivery of LDL-containing material toward a lysosome. *Cell 1, Slide 14*

WHY: The cell can keep the cargo but reuse the receptor, making repeated uptake efficient.

Exocytosis & membrane trafficking

Exocytosis [vesicle release to the outside] is the release of vesicle contents into extracellular space. *Cell 1, Slide 15*

WHY: A vesicle can fuse with the plasma membrane and empty its contents without rupturing the entire cell.

In many cells, exocytosis is triggered by a transient rise in cytosolic **Ca^{2+}** . *Cell 1, Slide 15*

WHY: Calcium provides a rapid "go" signal for vesicles that are ready to fuse.

The continuous movement and recycling of membrane is called **membrane trafficking**. The slide also links this trafficking to reducing blood lipid levels. *Cell 1, Slide 15*

WHY: Endocytosis removes membrane into the cell while exocytosis returns membrane to the surface, so the cell continuously recycles membrane.

Textbook supplement

The textbook separates transmembrane transport into **diffusion**, **channels**, **carriers**, and **pumps**. Channels form selective pores; carriers bind molecules and change conformation; pumps use ATP to move solutes against gradients.

Textbook, Ch. 2

WHY: These are different physical strategies for moving material across the same lipid barrier.

Water commonly crosses through **aquaporins** [water-channel proteins]. *Textbook, Ch. 2*

WHY: A dedicated water channel allows rapid, controlled water movement through an otherwise hydrophobic membrane.

The textbook adds **transcytosis** [vesicular transport across a cell]: some pinocytotic/endosomal vesicles move to the opposite cell surface and release contents there. *Textbook, Ch. 2*

WHY: This is a way to move bulk material from one side of a cell to the other without passing through the cytosol.

After receptor-mediated endocytosis, coated vesicles lose clathrin and enter the **endosomal compartment**; ligands may be degraded, receptors may recycle, or material may undergo transcytosis. *Textbook, Ch. 2*

WHY: Endosomes act as a sorting station for incoming cargo.

The textbook distinguishes **constitutive secretion** [continuous release] from **regulated secretion** [release after a stimulus]. *Textbook, Ch. 2*

WHY: Some products are shipped as soon as they are made, while others are stored until the cell receives a release signal.

Section Summary Table

Concept	Must know	Exam connection
Passive transport	Down gradient; no direct ATP use	Simple diffusion, channels, carriers/osmosis.
Active transport	Energy-dependent; can oppose gradient	Pumps; primary/secondary transport.
Phagocytosis	Large particles -> phagosome	Bacteria/dead cells.
Pinocytosis	Extracellular fluid -> small vesicles	~80 nm vesicles on slide.
Receptor-mediated	Ligand + receptor + clathrin-coated pit	Selective uptake; LDL is classic slide example.
Exocytosis	Vesicle fusion releases cargo	Often triggered by cytosolic Ca ²⁺ .
Membrane trafficking	Continuous membrane movement/recycling	Links endocytosis and exocytosis.

4. Cell Signaling: Hydrophilic vs Hydrophobic Signals

Primary slides: Cell 1, Slides 16-17. Textbook supplement: Chapter 2, Signal Reception & Transduction.

Slide-derived core

Hydrophilic signaling molecules use **signal transduction** [conversion of an outside signal into an intracellular response]. *Cell 1, Slide 16*

WHY: Water-soluble signals generally cannot cross the membrane's hydrophobic core, so they pass the message through a membrane receptor.

Effector proteins are commonly ion channels or enzymes that generate many **second messengers** [small intracellular signal-relay molecules]. *Cell 1, Slide 16*

WHY: One activated receptor can create many internal messengers, amplifying the signal.

Second messengers listed on the slide are **1,2-diacylglycerol (DAG)**, **cyclic adenosine monophosphate (cAMP)**, and **inositol 1,4,5-triphosphate (IP3)**. *Cell 1, Slide 16*

WHY: These molecules carry the signal deeper into the cell and activate downstream responses.

Hydrophobic signaling molecules include small steroid and thyroid hormones; they bind reversibly to carrier proteins in plasma for transport through the body and use **intracellular receptor proteins**. *Cell 1, Slide 17*

WHY: Lipid-soluble signals can cross the plasma membrane, but they need carriers to travel efficiently in watery blood plasma.

Textbook supplement

The textbook categorizes signaling by distance: **endocrine** (blood-borne hormones), **paracrine** (local diffusion), **synaptic** (neurotransmitter across a synapse), **autocrine** (signal acts on the producing cell), and **juxtacrine** (direct contact).

Textbook, Ch. 2

WHY: The category tells you how far the signal travels and how it reaches its target.

Hydrophilic ligands can use **channel-linked receptors**, **enzymatic receptors**, or **G-protein-coupled receptors (GPCRs)**. *Textbook, Ch. 2*

WHY: Different receptor designs trigger different intracellular responses, but all solve the same problem: transmitting an outside message across the membrane.

The textbook calls the extracellular ligand a **first messenger** and explains that intracellular second messengers amplify the signal and activate enzyme cascades, often involving kinases. *Textbook, Ch. 2*

WHY: The first messenger starts the message; the second messengers multiply and distribute it.

For steroid hormones, the textbook describes ligand-receptor complexes that can move to the nucleus and bind specific DNA sequences, generally altering transcription. *Textbook, Ch. 2*

WHY: A lipid-soluble signal can directly influence gene expression once it binds its intracellular receptor.

Section Summary Table

Concept	Must know	Exam connection
Hydrophilic signal	Membrane receptor -> transduction	Second messengers: DAG, cAMP, IP3.
Hydrophobic signal	Carrier in plasma; intracellular receptor	Steroid and thyroid hormones.
First vs second messenger	Ligand outside vs relay molecule inside	Signal amplification.
Receptor classes	Channel-linked, enzymatic, GPCR	All convert extracellular information into a cell response.

5. Nucleus, Nuclear Envelope/Pores, Lamina, Nucleolus & Chromatin

Primary slides: Cell 1, Slides 18-25. No matching nucleus chapter was present in the provided textbook PDF; this section therefore remains slide-based.

Nucleus

The **nucleus** [DNA-containing control compartment] serves as the blueprint for cell structures and activities. *Cell 1, Slide 18*

WHY: DNA stores the instructions that direct what the cell builds and does.

The nucleus is the site of **DNA replication** and transcription into precursor RNA molecules, and it contains enzymes involved in replication and repair of newly synthesized DNA. *Cell 1, Slide 18*

WHY: The nucleus both copies genetic information and converts it into RNA instructions.

The nucleus is enclosed by the **nuclear envelope** [double-membrane boundary around the nucleus] and contains the nuclear lamina, nucleolus, and chromatin. *Cell 1, Slide 18*

WHY: These structures organize the genome, support the nucleus, and produce ribosomal components.

The nucleus diagram labels **nuclear pores**, nuclear envelope, nucleolus, chromatin, ribosomes, and rough ER. It lists two broad nuclear functions: **cellular regulation** through genetic material and **production** of ribosomal subunits in the nucleolus for export to the cytoplasm. *Cell 1, Slide 19*

WHY: The diagram connects nuclear structure to its two big jobs: control of cell activity and preparation of ribosome components.

The nuclear-envelope slide visually emphasizes the **nuclear envelope** and **nuclear pores** in electron micrographs. *Cell 1, Slide 20*

WHY: Pores are the gateways through an otherwise continuous nuclear boundary.

Nuclear lamina

The **nuclear lamina** [protein mesh beneath the inner nuclear membrane] is a lattice-like network containing **lamins**. Lamins attach chromatin to the inner nuclear-envelope membrane. *Cell 1, Slide 21*

WHY: The lamina works like an internal supporting net that connects nuclear shape with genome organization.

Functions listed for the nuclear lamina are maintenance of nuclear shape, spatial organization of nuclear pores, regulation of transcription, anchoring of interphase heterochromatin, and DNA replication. *Cell 1, Slide 21*

WHY: The lamina is not just a skeleton; where DNA and pores are positioned can influence nuclear function.

Nucleolus

The **nucleolus** [non-membrane nuclear region that makes ribosomal components] is responsible for **rRNA synthesis** and **ribosome assembly**. *Cell 1, Slide 22*

WHY: Ribosomes need rRNA, so the cell concentrates rRNA production and early assembly in one nuclear region.

The **granular zone** is peripheral and contains ribosomal precursor particles in different stages of assembly. *Cell 1, Slide 22*

WHY: This is where developing ribosomal particles become more mature.

The **fibrillar zone** is centrally located and contains ribonuclear-protein fibrils; the **fibrillar center** contains DNA that is not being transcribed. *Cell 1, Slide 22*

WHY: Different nucleolar regions correspond to different stages or components of ribosome production.

Chromatin

Chromatin [DNA packaged with proteins] consists of DNA, histone proteins, and nonhistone proteins. *Cell 1, Slide 23*

WHY: DNA is not loose in the nucleus; proteins package it and help control its use.

The slide lists **B-DNA**, **Z-DNA**, and **A-DNA** forms. *Cell 1, Slide 23*

WHY: DNA can adopt more than one structural conformation.

Histones are positively charged proteins rich in lysine and arginine. *Cell 1, Slide 23*

WHY: Positive charge helps histones associate with negatively charged DNA.

Nonhistone proteins include enzymes involved in replication, transcription, DNA repair, and regulation; the slide describes them as acidic or neutral proteins. *Cell 1, Slide 23*

WHY: These proteins carry out much of the active work performed on DNA.

Heterochromatin [tightly packed, transcriptionally inactive chromatin] is highly condensed and is listed as about **10%** of total chromatin. *Cell 1, Slide 24*

WHY: Tightly packed DNA is difficult for transcription machinery to access.

Euchromatin [more open, transcriptionally active chromatin] is a more extended form of DNA and is listed as about **90%** of total chromatin; the slide states that only about **10%** of total chromatin is actively transcribed. *Cell 1, Slide 24*

WHY: Open DNA is more accessible, but even accessible DNA is not all being transcribed at once.

EXAM CHECK - In which nuclear structure is DNA actively transcribed to rRNA?

Correct answer: Nucleolus.

WHY: The nucleolus is the slide-defined site of rRNA synthesis and ribosome assembly.

Cell 1, Slide 25

Section Summary Table

Concept	Must know	Exam connection
Nucleus	DNA replication, RNA transcription, DNA repair	Genetic control center.
Nuclear envelope/pores	Boundary + exchange gateways	Connect nuclear and cytoplasmic compartments.
Nuclear lamina	Lamins; shape + chromatin/pore organization	Links structure and genome organization.
Nucleolus	rRNA synthesis + ribosome assembly	Correct location for rRNA transcription.
Heterochromatin	Condensed, inactive; ~10% on slide	Closed/less accessible.
Euchromatin	Extended, active; ~90% on slide	Open/more accessible.

6. Cytoplasm, Cytoplasmic Inclusions & Organelle Classification

Primary slides: Cell 2, Slides 1-4. Textbook supplement: Chapter 2, Cytoplasmic Organelles + Inclusions.

Slide-derived core

The **cytoplasm** is described as a homogeneous colloid with a pH of approximately **7.35**. *Cell 2, Slide 2*

WHY: A near-neutral fluid environment allows most cellular biochemical reactions to function properly.

The cytoplasm is about **70% water** and also contains salts, amino acids, enzymes, vitamins, hormones, gases, metabolites, and waste products. *Cell 2, Slide 2*

WHY: Water acts as the solvent in which the cell's molecules move and react.

High protein content together with **microfilaments** contributes to **sol-to-gel transformation** of cytoplasm. *Cell 2, Slide 2*

WHY: Changing protein and filament organization can change how fluid or firm the cytoplasm behaves.

The fluid cytoplasm, or **cytosol**, bathes metabolically active structures called organelles. *Cell 2, Slide 2*

WHY: Organelles operate inside a shared fluid environment that supplies substrates and carries products.

Cytoplasmic inclusions listed are **lipid droplets, glycogen, melanin, lipofuscin**, crystals of various types, and **hemosiderin**. *Cell 2, Slide 3*

WHY: Inclusions are stored or accumulated materials rather than the cell's main metabolic machinery.

Membranous organelles listed are mitochondria, smooth and rough ER, Golgi apparatus, lysosomes, peroxisomes, vacuoles, and vesicles. *Cell 2, Slide 4*

WHY: Membrane-bound compartments can maintain a specialized chemical environment separate from the cytosol.

Non-membranous structures listed are ribosomes, centrioles, microtubules, inclusions, cilia/flagella, and microvilli. *Cell 2, Slide 4*

WHY: These structures perform specialized jobs without being enclosed by their own membrane compartment.

Textbook supplement

The textbook adds that organelles may be membranous or nonmembranous protein complexes and that many are positioned by movements along the cytoskeleton. *Textbook, Ch. 2*

WHY: The cytoskeleton is both a structural framework and an intracellular transport system.

The textbook states that oxygen, CO₂, ions, small substrates, metabolites, and waste products diffuse through cytoplasm, either freely or bound to proteins. *Textbook, Ch. 2*

WHY: Cytosol is the exchange medium connecting organelles to the rest of the cell.

Textbook inclusion material emphasizes lipid droplets, glycogen granules, pigment granules, and residual bodies/lipofuscin as common nonliving cytoplasmic inclusions. *Textbook, Ch. 2*

WHY: Inclusions often represent stored fuel, pigment, or indigestible remnants.

The textbook summary identifies **cilia** as short numerous extensions that move material, a **flagellum** as a long singular extension that propels sperm, and **microvilli** as membrane folds supported by microfilaments that increase surface area. *Textbook, Ch. 2 summary table*

WHY: These surface specializations either move material/cells or increase the membrane area available for absorption.

Section Summary Table

Concept	Must know	Exam connection
Cytosol	~70% water; pH ~7.35 on slide	Fluid medium for cell chemistry.
Inclusions	Lipid, glycogen, pigments, crystals	Storage/accumulation rather than active organelles.
Membranous organelles	ER, Golgi, lysosome, peroxisome, mitochondria, vesicles	Compartmentalized chemistry.

Concept	Must know	Exam connection
Non-membranous structures	Ribosomes, centrioles, cytoskeleton, etc.	No separate organelle membrane.
Surface specializations	Cilia, flagellum, microvilli	Movement or increased surface area.

7. Ribosomes, Polysomes & Protein Targeting

Primary slides: Cell 2, Slides 5-6. Textbook supplement: Chapter 2, Ribosomes.

Slide-derived core

Ribosomes [rRNA-protein machines that synthesize proteins] are composed of rRNA and protein and have a large **60S** subunit and small **40S** subunit. *Cell 2, Slide 5*

WHY: The two subunits assemble on mRNA to translate genetic information into a polypeptide.

A **polysome / polyribosome** [multiple ribosomes translating one mRNA] consists of one mRNA being translated by several ribosomes at the same time. Ribosomes move along mRNA from the **5' end toward the 3' end**. *Cell 2, Slide 5*

WHY: Many ribosomes on one message let the cell produce multiple copies of a protein efficiently.

The free-vs-bound polyribosome figure shows **free polyribosomes** producing proteins of the cytosol/cytoskeleton and specific proteins imported into mitochondria, peroxisomes, and nucleus. *Cell 2, Slide 6*

WHY: Free ribosomes generally make proteins that remain in the cytosol or are imported into certain organelles after translation.

The same figure shows **ER-bound polyribosomes** sending proteins into the Golgi-processing/sorting pathway, leading to secretory vesicles, lysosomes, proteins secreted from the cell, and proteins of the cell membrane. *Cell 2, Slide 6*

WHY: Entry into RER routes a protein into the endomembrane/secretory system.

The slide figure also shows misfolded/denatured proteins being conjugated to **ubiquitin** and sent to the **proteasome** for protein degradation. *Cell 2, Slide 6*

WHY: The cell tags defective proteins so they can be selectively destroyed instead of accumulating.

Textbook supplement

The textbook describes ribosomes as macromolecular machines about **20 x 30 nm** that assemble amino acids in the sequence specified by mRNA. *Textbook, Ch. 2*

WHY: A ribosome is physically small but contains the machinery that converts an RNA code into protein.

Ribosomal rRNA is not only structural; the textbook states that it positions tRNAs and catalyzes peptide-bond formation.

Textbook, Ch. 2

WHY: The RNA core of the ribosome performs essential catalytic work rather than merely holding proteins together.

The textbook notes that intensely basophilic cytoplasmic regions indicate abundant RNA and active protein synthesis.

Textbook, Ch. 2

WHY: RNA phosphate groups bind basic stains strongly, so protein-producing regions often appear basophilic.

Section Summary Table

Concept	Must know	Exam connection
Ribosome	40S + 60S; rRNA + protein	Translation.
Polysome	Multiple ribosomes on one mRNA	High-output protein production.
Free polysome	Cytosolic/cytoskeletal + selected imported proteins	Not in secretory pathway.
ER-bound polysome	Secreted, membrane, lysosomal-pathway proteins	Feeds RER -> Golgi route.
Ubiquitin-proteasome	Tag defective protein -> degradation	Protein quality control.

8. Rough & Smooth Endoplasmic Reticulum + Medical Applications

Primary slides: Cell 2, Slides 7-12. Textbook supplement: Chapter 2, Endoplasmic Reticulum.

Rough ER

Rough endoplasmic reticulum (RER) [ribosome-studded membrane network] has a single lipid bilayer and is organized into stacks called **cisternae** [flattened membrane sacs]. *Cell 2, Slide 7*

WHY: The flattened sacs create a large membrane/lumen system where newly made proteins can enter and be processed.

RER synthesizes proteins destined for the Golgi apparatus, secretion, plasma membrane, and lysosomes. The slide gives pancreatic acinar cells as an example. *Cell 2, Slide 7*

WHY: A cell that exports many proteins needs abundant RER because secretory proteins enter the ER early in synthesis.

The RER-targeting diagram shows an initial **signal peptide**, binding of **signal recognition particle (SRP)**, docking to an SRP receptor and ribosome/translocator complex, SRP release, signal-peptide removal by signal peptidase, and continued growth of the polypeptide into the RER cisterna. *Cell 2, Slide 8*

WHY: The signal peptide behaves like an address label that directs the translating ribosome to the RER.

Slide 9 is a histologic visual reinforcement slide associated with the preceding RER topic; it adds no new written facts to the deck. *Cell 2, Slide 9*

WHY: The image is meant to connect the organelle concept to how RER-rich cells appear in tissue.

Smooth ER

Smooth endoplasmic reticulum (SER) [ER lacking attached ribosomes] is a network of membranous sacs, vesicles, and tubules continuous with RER. *Cell 2, Slide 10*

WHY: RER and SER are regions of one connected membrane system, but each region specializes in different tasks.

SER contains enzymes involved in synthesis of **phospholipids, triglycerides, and sterols**. *Cell 2, Slide 10*

WHY: Without ribosomes, SER is specialized for lipid-related chemistry rather than ribosome-driven protein synthesis.

SER functions listed are detoxification reactions (**hydroxylation** and **conjugation**), glycogen degradation and gluconeogenesis, reactions in lipid metabolism, and sequestration/release of Ca^{2+} . *Cell 2, Slide 11*

WHY: SER acts as a chemical-processing and storage network for lipids, drugs/toxins, carbohydrate metabolism, and calcium.

Medical applications

Jaundice [yellow discoloration caused by bilirubin buildup] is described as accumulation of bilirubin, which is normally metabolized by SER enzymes in liver cells and excreted as bile. *Cell 2, Slide 12*

WHY: If bilirubin handling does not keep pace, the yellow pigment accumulates in tissues.

Osteogenesis imperfecta [fragile-bone disorder] is presented as defective procollagen synthesis/secretion producing very weak bone tissue, linked on the slide to defects in RER and **ER-associated degradation (ERAD)**. *Cell 2, Slide 12*

WHY: Bone strength depends heavily on correctly produced collagen; defective processing weakens the structural framework.

Textbook supplement

The textbook describes the ER as a continuous network extending from the nuclear surface through the cytoplasm and enclosing interconnected cisternae. *Textbook, Ch. 2*

WHY: The ER functions as one continuous intracellular membrane system rather than isolated sacs.

New proteins entering RER undergo folding and quality control; improperly folded proteins can be targeted to ubiquitin/proteasomal degradation. *Textbook, Ch. 2*

WHY: The RER acts as both a protein factory and an early quality-control checkpoint.

The textbook reinforces that SER is especially important in cells active in lipid/steroid metabolism, detoxification, glycogen-related metabolism, and calcium handling. *Textbook, Ch. 2*

WHY: Different cell types expand the SER function they need most.

Section Summary Table

Concept	Must know	Exam connection
RER structure	Ribosomes + cisternae	Protein production for secretory/endomembrane pathway.
RER targeting	Signal peptide -> SRP -> receptor/translocator	Co-translational entry into RER.
SER structure	Tubules/sacs; no ribosomes	Continuous with RER.
SER functions	Lipids, detox, glycogen/gluconeogenesis, Ca ²⁺	Know the function cluster.
Jaundice	Bilirubin accumulation; liver SER metabolism on slide	SER medical application.
Osteogenesis imperfecta	Defective procollagen/RER-ERAD framing	Protein-processing defect -> weak bone.

9. Golgi Apparatus, COP Coats & I-Cell Disease

Primary slides: Cell 2, Slides 13-16. Textbook supplement: Chapter 2, Golgi Apparatus.

Golgi structure and functions

The **Golgi apparatus** [cellular protein/lipid processing and sorting station] consists of disc-shaped smooth cisternae (**dictyosomes**) and membrane-bound vesicles. *Cell 2, Slide 13*

WHY: Flattened sacs and transport vesicles create a stepwise processing-and-shipping system.

The **cis (forming) face** is associated with the RER, whereas the **trans (maturing) face** is often oriented toward the plasma membrane. *Cell 2, Slide 13*

WHY: Cargo arrives from RER at the cis side and exits after processing from the trans side.

Golgi functions emphasized are **glycosylation, phosphorylation, sulfation**, and synthesis/concentration/storage of secretory products. *Cell 2, Slide 13*

WHY: Chemical modification can mature a protein and also help determine where it should be sent.

The coat-direction slide states: **COP-II = forward movement** and **COP-I = retrograde movement**. *Cell 2, Slide 14*

WHY: Forward transport advances cargo from ER toward Golgi, while retrograde transport retrieves/recycles material in the opposite direction.

The detailed Golgi diagram shows this sequence: nucleus makes mRNA/tRNA/rRNA; RER performs translation, segregation, signal removal, and beginning of glycosylation; cis Golgi phosphorylates lysosomal glycoproteins; medial Golgi removes mannose and adds N-acetyl-glucosamine, then adds galactose; the trans Golgi performs packing, condensation, sorting, final proteolysis, and specific distribution. *Cell 2, Slide 15*

WHY: The figure presents the secretory pathway as an assembly line where each compartment performs a different processing step.

The trans Golgi network in the diagram sorts cargo toward membrane proteins, secretory granules/secretion, or lysosomes. *Cell 2, Slide 15*

WHY: The trans side is the final shipping hub where processed cargo is assigned a destination.

Medical application

I-cell disease [lysosomal enzyme-targeting disorder] is listed as autosomal recessive and caused by deficiency of **N-acetylglucosamine-phosphotransferase**, which normally phosphorylates mannose residues to form **mannose-6-phosphate (M6P)** on N-linked glycoproteins in the Golgi. *Cell 2, Slide 16*

WHY: M6P acts like a lysosomal address tag; without proper tagging, lysosomal enzymes are mis-sorted.

Textbook supplement

The textbook reinforces that RER-derived cargo enters the cis Golgi in membrane carriers, moves through sequential Golgi compartments, and is sorted at the trans Golgi network. *Textbook, Ch. 2*

WHY: Physical direction through the Golgi matches functional maturation of cargo.

The textbook states that **M6P** marks lysosomal enzymes and that receptors for M6P-containing proteins divert them from the general secretory pathway into lysosome-forming vesicles. *Textbook, Ch. 2*

WHY: The M6P system is the sorting mechanism that explains why a tagging defect causes lysosomal targeting failure.

The textbook describes **secretory granules** as Golgi-derived vesicles that can store concentrated product until regulated secretion is triggered. *Textbook, Ch. 2*

WHY: Some cells make secretion in advance, package it densely, and wait for a release signal.

Section Summary Table

Concept	Must know	Exam connection
Cis Golgi	Receiving/forming face near RER	Entry side.

Concept	Must know	Exam connection
Trans Golgi	Maturing/sorting side	Exit/shipping side.
Golgi modifications	Glycosylation, phosphorylation, sulfation	Cargo maturation + targeting.
COP-II	Forward movement	ER -> Golgi direction.
COP-I	Retrograde movement	Retrieval/recycling direction.
I-cell disease	Defective phosphotransferase -> defective M6P tagging	Lysosomal enzyme mis-targeting.

10. Lysosomes, Lysosomal Storage Diseases & Residual Bodies

Primary slides: Cell 2, Slides 17-19. Textbook supplement: Chapter 2, Lysosomes.

Lysosome basics

Lysosomes [acidic intracellular digestive organelles] are spherical membrane-enclosed organelles containing enzymes required for intracellular digestion. *Cell 2, Slide 17*

WHY: Digestive enzymes are kept inside a membrane compartment so they can break down cargo without digesting the cytoplasm around them.

The slides distinguish **primary lysosomes** and **secondary lysosomes**. *Cell 2, Slide 17*

WHY: This distinction reflects lysosomal digestive compartments before vs after fusion with material to be digested.

All lysosomal enzymes are **acid hydrolases** [digestive enzymes that work best in acidic conditions], with optimal activity at approximately **pH 5.0**. *Cell 2, Slide 17*

WHY: The acidic lysosomal lumen activates digestive enzymes, while the near-neutral cytosol makes escaped enzymes less active.

Lysosomal storage disease table - memorize the pairings

Hurler syndrome (MPS I) [lysosomal storage disease] - faulty enzyme: **alpha-L-iduronidase**; main organs affected: skeleton and nervous system. *Cell 2, Slide 18*

WHY: A missing lysosomal enzyme causes material normally degraded by that enzyme to accumulate, so tissues with heavy burden become dysfunctional.

McArdle syndrome [glycogen-metabolism disorder listed in the slide table] - faulty enzyme: **muscle phosphorylase**; main organ affected: skeletal muscle. *Cell 2, Slide 18*

WHY: The slide's key exam pairing is disease -> enzyme -> affected tissue.

Tay-Sachs [lysosomal storage disease] - faulty enzyme: **GM2 gangliosidase**; main organ affected: nervous system. *Cell 2, Slide 18*

WHY: Failure to break down the relevant substrate particularly injures the nervous system in this slide pairing.

Gaucher disease [lysosomal storage disease] - faulty enzyme: **glucocerebrosidase**; main organs affected: liver and spleen. *Cell 2, Slide 18*

WHY: The high-yield task is recognizing the enzyme and organ association shown in the table.

I-cell disease - faulty enzyme: **phosphotransferase for M6P formation**; main organs affected: skeleton and nervous system. *Cell 2, Slide 18*

WHY: Unlike a single digestive-enzyme deficiency, this defect disrupts lysosomal enzyme targeting at the Golgi level.

Residual bodies [vesicles containing indigestible remnants after lysosomal digestion] are shown as the end-stage leftover material after lysosomal processing. *Cell 2, Slide 19*

WHY: Not everything the lysosome receives can be broken down completely, so indigestible remnants may remain enclosed.

Textbook supplement

The textbook describes fusion of a primary lysosome with a phagosome/endosomal compartment to create an active **secondary lysosome / heterolysosome**. *Textbook, Ch. 2*

WHY: Digestion becomes active when enzyme-containing lysosomes meet the material that needs to be broken down.

The textbook adds **autophagy** [lysosomal digestion of the cell's own unneeded components]: organelles can be enclosed in an autophagosome that fuses with a lysosome. *Textbook, Ch. 2*

WHY: Autophagy lets the cell recycle worn-out internal parts instead of only digesting material from outside.

Indigestible material may remain in residual bodies and can accumulate as **lipofuscin** in long-lived cells. *Textbook, Ch. 2*

WHY: Long-lived cells have more time to accumulate material that cannot be fully degraded.

Section Summary Table

Concept	Must know	Exam connection
Lysosome	Membrane-bound intracellular digestion	Acid hydrolases.
Optimal pH	~5.0	Acidic lumen activates enzymes.
Primary -> secondary	Fusion with material to digest	Creates active digestive compartment.
Residual body	Indigestible remnants	May persist/accumulate.
M6P system	Targets lysosomal enzymes	I-cell disease disrupts this.
Autophagy	Digestion/recycling of cell's own components	Internal quality-control pathway.

11. Peroxisomes & Proteasomes

Primary slides: Cell 2, Slides 20-21. Textbook supplement: Chapter 2, Peroxisomes + Proteasomes.

Peroxisomes

Peroxisomes [small single-membrane oxidative organelles] are a heterogeneous group of small spherical organelles surrounded by one membrane. *Cell 2, Slide 20*

WHY: A dedicated membrane compartment safely concentrates oxidation reactions.

Slide functions are synthesis and degradation of **hydrogen peroxide (H₂O₂)**, oxidation of **very-long-chain fatty acids (> C₂₄)**, phospholipid exchange, and bile-acid synthesis. *Cell 2, Slide 20*

WHY: Peroxisomes specialize in oxidation and lipid-related reactions that complement other organelles.

Peroxisome deficiency causes failure to oxidize very-long-chain fatty acids and accumulation of bile-acid precursors. *Cell 2, Slide 20*

WHY: When the required organelle pathway is defective, the substrates normally handled by it build up.

Disorders listed are **Zellweger (cerebrohepatorenal) syndrome** [peroxisomal disorder] and **neonatal adrenoleukodystrophy** [peroxisomal lipid-transport/metabolism disorder]. *Cell 2, Slide 20*

WHY: These disease names are high-yield examples linking clinical disease to defective peroxisomal function.

Proteasomes

Proteasomes [non-membranous protein-destruction complexes] are non-membranous protein complexes that degrade denatured or nonfunctional polypeptides. *Cell 2, Slide 21*

WHY: Proteasomes are a protein quality-control disposal system.

The slide associates **Alzheimer disease** and **Huntington disease** with protein aggregates due to failure of proteasomes. *Cell 2, Slide 21*

WHY: The teaching point is that inadequate removal of abnormal proteins allows damaging aggregates to accumulate.

Textbook supplement

The textbook states that peroxisomal oxidases generate H₂O₂ and **catalase** breaks H₂O₂ down to water and oxygen.

Textbook, Ch. 2

WHY: The organelle creates a potentially harmful oxidant and contains an enzyme that neutralizes it in the same compartment.

Peroxisomal proteins are synthesized on free polyribosomes and imported using targeting sequences recognized by peroxisomal receptors. *Textbook, Ch. 2*

WHY: Peroxisomes do not use RER-bound ribosomes for their own enzymes; their proteins are made in the cytosol and imported.

The textbook describes proteasomal degradation as ATP-dependent and targeted by **ubiquitin** tagging; proteasomes remove defective proteins and proteins no longer needed by the cell. *Textbook, Ch. 2*

WHY: Ubiquitin works like a disposal label, allowing selective degradation instead of random protein destruction.

Section Summary Table

Concept	Must know	Exam connection
Peroxisome membrane	Single membrane	Oxidative compartment.
VLCFA oxidation	Very-long-chain fatty acids >C ₂₄	Defect -> accumulation.
H ₂ O ₂ handling	Oxidation + catalase detoxification	Contain reactive chemistry.
Proteasome	Non-membranous protein degradation	Denatured/nonfunctional proteins.
Ubiquitin	Protein-degradation tag	Targets proteins to proteasome.

12. Mitochondria, Cristae, Matrix, mtDNA & MERRF

Primary slides: Cell 2, Slides 22-25 and 41. Textbook supplement: Chapter 2, Mitochondria.

Structure & energy production

Mitochondria [ATP-producing double-membrane organelles] synthesize ATP and contain double-stranded circular DNA and proteins. They have two membranes. *Cell 2, Slide 22*

WHY: Mitochondria combine their own genetic material with a highly specialized membrane system for energy production.

The **outer mitochondrial membrane** is smooth, continuous, and highly permeable because of abundant **porins** [channel proteins]. *Cell 2, Slide 22*

WHY: The outer membrane acts as a relatively open gateway to the organelle.

The **intermembrane compartment** [space between outer and inner mitochondrial membranes] contains enzymes that use ATP to phosphorylate other nucleotides, including creatine phosphokinase and adenylate kinase as listed in the slide. *Cell 2, Slide 22*

WHY: This compartment participates in energy/nucleotide handling between the two mitochondrial membranes.

The **inner mitochondrial membrane** is impermeable to most small ions (Na⁺, K⁺, H⁺) and small molecules (ATP, ADP, pyruvate), has high **cardiolipin** content, requires carnitine for fatty-acid transport, and forms infoldings called **cristae**. *Cell 2, Slide 23*

WHY: Tight control of movement across the inner membrane is essential because energy production depends on maintaining a controlled electrochemical gradient.

Cristae contain enzymes for **electron transport** and **oxidative phosphorylation**. *Cell 2, Slide 23*

WHY: Folding the inner membrane creates more surface area for the machinery that makes ATP.

The **mitochondrial matrix** [innermost mitochondrial compartment] contains enzymes that oxidize pyruvate, amino acids, and fatty acids and generate reduced **NADH** and **FADH** for the electron-transport chain and energy generation. *Cell 2, Slide 23*

WHY: The matrix extracts high-energy electrons from fuels, and the inner membrane uses those electrons to drive ATP production.

Two cristae types are shown: **plate-like cristae** in most cells and **tubular cristae** in steroid-secreting cells such as adrenal and gonadal cells. *Cell 2, Slide 24*

WHY: Crista shape varies with the cell's specialized mitochondrial workload.

Mitochondrial genetics & disease

The slide states that **mitochondrial DNA is inherited from the mother**, producing maternal transmission of diseases of energy metabolism. *Cell 2, Slide 25*

WHY: The embryo receives its mitochondria primarily from the oocyte, creating a characteristic inheritance pattern.

Intramitochondrial granules contain calcium and magnesium. *Cell 2, Slide 25*

WHY: These granules represent stored mineral ions inside the mitochondrial matrix.

MERRF - myoclonic epilepsy with ragged-red fibers [mitochondrial disorder on the slide] is linked to mitochondrial DNA with a mutated gene for lysine-tRNA in skeletal muscle and defective synthesis of respiratory-chain protein. *Cell 2, Slide 25*

WHY: If mitochondrial protein synthesis is defective, the respiratory chain cannot work normally and ATP production suffers.

Textbook supplement

The textbook describes mitochondria as dynamic organelles that grow and divide by fission of preexisting mitochondria; mitochondrial number varies with cellular energy demand. *Textbook, Ch. 2*

WHY: Cells can expand mitochondrial capacity when they need more ATP production.

The textbook also connects mitochondria with regulated **apoptosis** [programmed cell death] through release of mitochondrial components. *Textbook, Ch. 2*

WHY: Mitochondria are not only energy organelles; they also participate in the internal cell-death program.

EXAM CHECK - Mitochondrial content is proportional to energy demand. Which cell type in the slide question should have the greatest mitochondrial content?

Correct answer: Cardiac muscle cells.

WHY: Cardiac muscle contracts continuously and therefore has a very high ATP requirement.

Cell 2, Slide 41

Section Summary Table

Concept	Must know	Exam connection
Outer membrane	Porin-rich; highly permeable	Entry boundary.
Inner membrane	Cardiolipin; selective; ETC/oxidative phosphorylation	Energy-conserving membrane.
Cristae	Inner-membrane folds	Increase functional membrane area.
Matrix	Fuel oxidation → NADH/FADH	Feeds electron-transport chain.
mtDNA	Maternal inheritance on slide	Mitochondrial-disease pedigree clue.
MERRF	Lysine-tRNA mutation; defective respiratory-chain protein synthesis	Mitochondrial translation/energy defect.

13. Cytoskeleton: Microtubules, Motors, MTOC, Chediak-Higashi & Centrioles

Primary slides: Cell 2, Slides 26-32. Textbook supplement: Chapter 2, The Cytoskeleton - Microtubules.

Microtubules

The **cytoskeleton** [internal protein framework] includes microtubules, microfilaments, and intermediate filaments. Slide 26 introduces this network visually. *Cell 2, Slide 26*

WHY: Cells need an internal support and transport system to maintain shape, move cargo, and divide.

Microtubules [hollow tubulin cylinders] participate in chromosomal movement during meiosis/mitosis, spindle formation, intracellular vesicle and organelle transport, and ciliary/flagellar movement. *Cell 2, Slide 27*

WHY: Microtubules are stiff, long-range structural tracks that are ideal for division and intracellular transport.

Microtubules are cylindrical unbranched tubules approximately **25 nm** in diameter with a **5-nm-thick wall**, made from alpha/beta-tubulin heterodimers and organized into **13 protofilaments**. *Cell 2, Slide 27*

WHY: Repeating tubulin subunits assemble into a strong hollow tube.

Motor proteins & organizing center

Kinesin [microtubule motor] moves vesicles and organelles **away from the cell center**. *Cell 2, Slide 28*

WHY: Kinesin acts like a molecular vehicle moving cargo along microtubule tracks toward the cell periphery.

Dynein [microtubule motor] moves cargo on microtubules **toward the cell center**. *Cell 2, Slide 28*

WHY: Dynein provides movement in the opposite direction, allowing two-way intracellular traffic.

Microtubule formation is directed by a **microtubule-organizing center (MTOC)** [site that nucleates/organizes microtubules] and is controlled by Ca^{2+} concentration and **microtubule-associated proteins (MAPs)** according to the slide. *Cell 2, Slide 29*

WHY: A central organizing site prevents microtubules from growing in a completely random pattern.

Chediak-Higashi syndrome

Chediak-Higashi syndrome [autosomal-recessive immunodeficiency with abnormal intracellular transport] is presented as a defect in microtubule polymerization and abnormal intracellular protein transport. *Cell 2, Slide 30*

WHY: Immune cells depend on cytoskeletal transport to move vesicles and digestive compartments to the right place.

The slide states that Chediak-Higashi syndrome causes delayed fusion of phagosomes with lysosomes in leukocytes. *Cell 2, Slide 30*

WHY: If vesicles cannot move and fuse efficiently, ingested microbes are not delivered promptly to lysosomal enzymes.

Centrioles

Centrioles [paired cylindrical microtubule structures] occur as a pair with long axes perpendicular to each other and contain **9 sets of microtubule triplets** arranged in a pinwheel. *Cell 2, Slide 31*

WHY: The regular nine-triplet pattern forms a stable organizing structure at the centrosome.

Slide 32 visually reinforces centriole organization in longitudinal and cross-sectional views and its relationship to the centrosomal microtubule-organizing region. *Cell 2, Slide 32*

WHY: Seeing both views helps translate the “nine triplets” description into a three-dimensional cylinder.

Textbook supplement

The textbook describes microtubules as polarized structures with dynamic assembly/disassembly and identifies the **centrosome** as the dominant MTOC in most cells. *Textbook, Ch. 2*

WHY: Dynamic behavior lets cells rapidly reorganize their microtubule network for transport or cell division.

The textbook reinforces that the paired centrioles are oriented at right angles and that each centriole contains nine microtubule triplets; centrosomes duplicate before mitosis and form spindle poles. *Textbook, Ch. 2*

WHY: Centrosome duplication ensures that the dividing cell can organize a bipolar mitotic spindle.

Section Summary Table

Concept	Must know	Exam connection
Microtubule	25 nm; alpha/beta tubulin; 13 protofilaments	Spindle, transport, cilia/flagella.
Kinesin	Away from cell center	One direction of cargo transport.
Dynein	Toward cell center	Opposite direction of cargo transport.
MTOC/centrosome	Organizes microtubules	Directional network + spindle poles.
Centriole	9 microtubule triplets; paired at right angles	Centrosomal organization.
Chediak-Higashi	Defective transport; delayed phagosome-lysosome fusion	Microtubule/vesicle trafficking disease example.

14. Microfilaments (Actin) & Intermediate Filaments

Primary slides: Cell 2, Slides 33-36. Textbook supplement: Chapter 2, Microfilaments + Intermediate Filaments.

Microfilaments / actin

Slide 33 visually introduces **actin filaments** / **microfilaments** in cells. *Cell 2, Slide 33*

WHY: The image emphasizes that actin forms filamentous networks rather than membrane-bound compartments.

Microfilaments are polymers of the protein **actin**. Actin exists as globular **G-actin** and filamentous **F-actin**. *Cell 2, Slide 34*

WHY: G-actin is the building block; F-actin is the assembled filament.

F-actin polymerizes into helically entwined actin chains. The chains can dissociate and reassemble with changes in Ca^{2+} and cAMP levels according to the slide. *Cell 2, Slide 34*

WHY: Actin is useful for movement because it can be remodeled quickly.

Intermediate filaments

Intermediate filaments [stable rope-like cytoskeletal filaments] are stable and confer increased mechanical stability to cell structure. *Cell 2, Slide 35*

WHY: They behave like reinforcing cables that help cells resist mechanical stress.

Vimentin is found in cells of mesenchymal origin and may help position the nucleus. *Cell 2, Slide 36*

WHY: Vimentin is a high-yield marker for mesenchymal-type cells and contributes to internal structural organization.

Desmin is found at Z-disks of skeletal muscle cells and helps ensure uniform tension distribution. *Cell 2, Slide 36*

WHY: Muscle cells need intermediate filaments to spread mechanical force across the cell.

Glial fibrillary acidic protein (GFAP) is characteristic of glial cells (astrocytes). *Cell 2, Slide 36*

WHY: Different cell lineages express characteristic intermediate-filament proteins.

Neurofilaments are formed by three distinct proteins and are present in neuronal cytoplasm. *Cell 2, Slide 36*

WHY: Neurons use a specialized intermediate-filament system to support their long processes.

Keratins occur in skin cells and contribute to resistance to friction and cell-to-cell adhesion. *Cell 2, Slide 36*

WHY: Epithelial cells are exposed to mechanical stress, so strong keratin networks reinforce them.

Textbook supplement

The textbook gives approximate cytoskeletal diameters: **microfilaments 5-7 nm**, **intermediate filaments 8-10 nm**, and microtubules about 25 nm. *Textbook, Ch. 2*

WHY: The names “micro,” “intermediate,” and “microtubule” correspond to distinct filament sizes and structures.

The textbook emphasizes actin's roles in cell-surface shape changes, microvilli, endocytosis, cell movement, organelle/vesicle transport with myosins, and the contractile ring during cell division. *Textbook, Ch. 2*

WHY: Actin is concentrated where the cell needs rapid shape change or force generation.

Intermediate filaments are the most stable cytoskeletal component and use different protein subunits in different cell types, making them useful indicators of cell lineage. *Textbook, Ch. 2*

WHY: A durable, cell-type-specific structural protein can both support the cell and identify where that cell came from.

Section Summary Table

Concept	Must know	Exam connection
G-actin	Globular monomer	Building block of F-actin.
F-actin	Filamentous helical polymer	Dynamic shape/movement functions.
Intermediate filaments	Stable; 8-10 nm in textbook	Mechanical strength.
Vimentin	Mesenchymal cells	Cell-lineage marker/support.

Concept	Must know	Exam connection
Desmin	Muscle Z-disks	Distributes tension.
GFAP / neurofilaments / keratin	Glia / neurons / skin epithelial cells	Know the cell-type pairings.

15. Apoptosis: Mitochondrial, Death-Receptor & Cytotoxic-T-Cell Pathways

Primary slides: Cell 2, Slides 37-40. Textbook supplement: mitochondrial role in apoptosis from Chapter 2.

Intrinsic mitochondrial pathway

Apoptosis [organized programmed cell death] can be triggered by cellular injury, DNA damage, or decreased hormonal stimulation, leading to inactivation of **Bcl-2** according to the slide. *Cell 2, Slide 37*

WHY: When survival signals are lost or damage is severe, the cell can activate a controlled self-destruction program.

Loss of Bcl-2 activity allows **cytochrome c** to leak from mitochondria into the cytoplasm and activate **caspases** [proteases that execute apoptosis]. *Cell 2, Slide 37*

WHY: Cytochrome c release is a mitochondrial “point of no return” signal that activates the destruction machinery.

Caspases activate proteases and endonucleases. *Cell 2, Slide 37*

WHY: These enzymes systematically dismantle cellular proteins and DNA.

The apoptosis diagram expands the intrinsic pathway with triggers including DNA damage, radiation, reactive oxygen species/toxins, misfolded proteins, and hypoxia; it shows p53 activation, BAX/BAK, cytochrome c, Bcl-2 control, initiator caspases, and executioner caspases. *Cell 2, Slide 38*

WHY: The diagram shows that many types of serious cellular stress converge on a common caspase execution pathway.

Extrinsic & cytotoxic-T-cell pathways

The slide-38 diagram shows an **extrinsic death-receptor pathway** involving Fas/FasL and TNF-alpha/TNFR, which activates initiator caspases and then executioner caspases. *Cell 2, Slide 38*

WHY: A death signal can arrive from outside the cell and activate the same execution machinery without first being triggered by internal damage.

The same diagram shows a **cytotoxic T-cell** pathway in which perforin and granzyme B contribute to caspase activation. *Cell 2, Slide 38*

WHY: Immune cells can deliberately trigger apoptosis in a target cell as a controlled way to eliminate it.

Executioner caspases produce nuclear fragmentation, cytoskeletal dispersion, and cytoplasmic blebbing, followed by formation of apoptotic bodies that expose ligands for macrophage receptors. *Cell 2, Slide 38*

WHY: The cell breaks itself into small, recognizable packages that can be cleared efficiently.

Morphology & inflammation

During apoptosis the dying cell shrinks, the cytoplasm becomes more eosinophilic, and the nucleus condenses and fragments in an organized manner. *Cell 2, Slide 39*

WHY: Apoptosis produces a compact, orderly cell-death morphology rather than explosive rupture.

Apoptotic bodies [small membrane-bound fragments of a dying cell] are removed by macrophages, and apoptosis is not followed by inflammation. *Cell 2, Slide 39*

WHY: Because the fragments remain packaged and are rapidly cleared, intracellular contents do not spill widely into surrounding tissue.

Exam application: HPV E6 and p53

EXAM CHECK - High-risk HPV E6 binds p53 and causes p53 degradation. What mechanism best explains how E6 promotes cervical cancer?

Correct answer: Prevents apoptosis.

WHY: p53 helps damaged cells undergo cell-cycle arrest or apoptosis; destroying p53 allows abnormal cells to survive instead of being eliminated.

Cell 2, Slide 40

Textbook supplement

The textbook also identifies mitochondria as participants in regulated apoptosis through release of mitochondrial components. *Textbook, Ch. 2*

WHY: This independently reinforces the slide's central idea that mitochondria are key regulators of the intrinsic death pathway.

Section Summary Table

Concept	Must know	Exam connection
Intrinsic pathway	Cell stress -> mitochondrial cytochrome c -> caspases	Bcl-2/BAX-BAK control shown in diagram.
Extrinsic pathway	Fas/TNFR death receptors -> caspases	Outside death signal.
Cytotoxic T cell	Perforin/granzyme B -> caspase pathway	Immune-triggered apoptosis.
Morphology	Shrinkage, eosinophilia, nuclear fragmentation, blebs	Orderly cell dismantling.
Clearance	Apoptotic bodies -> macrophages	No inflammatory response on slide.
HPV E6	p53 degradation	Prevents apoptosis in slide question.

16. Integrated High-Yield Pathways & Rapid Review

Integrates the uploaded slide decks and textbook supplementation; no outside material added.

Integrated pathways

Secreted-protein pathway: mRNA translation begins -> signal peptide/SRP targets RER -> protein enters RER -> COP-II forward transport -> cis Golgi -> sequential Golgi modification -> trans Golgi sorting -> secretory vesicle -> exocytosis.

WHY: This is the cell's production-and-shipping line for proteins destined to leave the cell or enter the endomembrane system.

Lysosomal-enzyme pathway: RER synthesis -> Golgi processing -> M6P tagging -> M6P-receptor sorting -> lysosome.

WHY: A lysosomal enzyme needs both correct synthesis and a correct "address label."

Receptor-mediated LDL pathway: LDL binds receptor -> clathrin-coated pit -> coated vesicle -> endosome -> receptor recycling + LDL-directed lysosomal pathway.

WHY: Selective uptake and receptor recycling let the cell repeatedly capture a specific extracellular molecule.

Intracellular digestion pathway: phagosome/pinocytotic vesicle -> lysosomal fusion -> acid-hydrolase digestion -> residual body if material is indigestible.

WHY: Endocytosis brings material in; lysosomes perform the breakdown.

Mitochondrial energy pathway: matrix oxidizes fuels -> NADH/FADH -> electron transport on inner membrane/cristae -> oxidative phosphorylation -> ATP.

WHY: Fuel oxidation loads electron carriers, and the inner membrane converts that energy into usable ATP.

Microtubule transport pathway: MTOC organizes microtubules; kinesin moves cargo away from the center and dynein moves cargo toward the center.

WHY: A polarized track plus motors in opposite directions creates two-way intracellular traffic.

Apoptosis pathway: severe stress/loss of survival control -> mitochondrial cytochrome c or death-receptor signal -> initiator/executioner caspases -> cell fragmentation -> macrophage clearance.

WHY: Different triggers converge on a common execution program that removes the cell cleanly.

Must-memorize contrasts

Section Summary Table

Concept	Must know	Exam connection
RER vs SER	RER: ribosomes/proteins; SER: no ribosomes/lipids-detox-Ca ²⁺	Do not mix protein vs lipid roles.
Cis vs trans Golgi	Cis receives; trans sorts/ships	Direction through Golgi.
COP-II vs COP-I	COP-II forward; COP-I retrograde	Classic coat-direction question.
Lysosome vs proteasome	Acidic membrane digestion vs non-membrane protein degradation	Different substrates and compartments.
Phagocytosis vs pinocytosis	Large particles vs extracellular fluid	Cell eating vs cell drinking.
Hydrophilic vs hydrophobic signals	Membrane receptor/2nd messenger vs intracellular receptor	Membrane permeability determines receptor location.
Kinesin vs dynein	Away from center vs toward center	Opposite microtubule motors.
Euchromatin vs heterochromatin	Extended/active vs condensed/inactive	Accessibility predicts transcription.
Actin vs IF vs microtubule	Dynamic thin filament vs stable reinforcement vs hollow transport/spindle tube	Know structure + function + size relationship.

17. Textbook Expansion - Differentiation, Membrane Structure & Membrane Dynamics

Expanded from the uploaded textbook, Chapter 2, pp. 17-20. These points are included because they are important textbook material not fully developed in the slides.

Cell differentiation beyond the slide table

The textbook traces specialized cells back to the **zygote** [the single cell formed at fertilization]. Early divisions produce **blastomeres** [early embryonic cells], and cells of the inner cell mass can give rise to the tissue types of the fetus.

Textbook, Ch. 2, p. 17

WHY: This explains why very different adult cells can share the same basic genome: they descend from common early cells but later use different gene programs.

During **differentiation** [cell specialization], cells predominantly express selected sets of genes, alter cytoplasmic activities, and usually change shape to become efficient at a particular tissue function. *Textbook, Ch. 2, p. 17*

WHY: Specialization is not just a change in appearance; it is a coordinated change in gene expression, organelles, proteins, and cell shape.

The same apparent cell type can vary with its microenvironment, and structurally similar cells can respond differently because they express different families of receptors for hormones or extracellular-matrix signals. *Textbook, Ch. 2, p. 17*

WHY: A cell's behavior depends not only on what it looks like, but also on which signals it can detect.

Membrane lipid organization and microscopy

The two leaflets of the plasma membrane have **asymmetric lipid composition**. The textbook example notes more phosphatidylcholine and sphingomyelin externally and more phosphatidylserine and phosphatidylethanolamine internally in red-cell membranes. *Textbook, Ch. 2, p. 18*

WHY: Different lipid mixtures on each side allow the inner and outer surfaces to perform different chemical and signaling functions.

After osmium fixation, membranes may show a **trilaminar appearance** on TEM: two electron-dense outer lines surrounding a lighter hydrophobic middle region. *Textbook, Ch. 2, pp. 18-19*

WHY: Osmium associates with polar membrane regions more than the fatty-acid interior, so the bilayer appears as dark-light-dark.

Integral proteins are embedded in the lipid bilayer and many are multipass transmembrane proteins; **peripheral proteins** are attached more loosely to a membrane surface and can often be removed without disrupting the bilayer.

Textbook, Ch. 2, pp. 18-20

WHY: The strength and location of protein-membrane association reflect what the protein needs to do.

Freeze-fracture studies split membranes through the hydrophobic center and reveal intramembranous particles representing many membrane proteins. *Textbook, Ch. 2, p. 20*

WHY: Splitting the bilayer exposes the otherwise hidden membrane interior and shows that proteins are embedded within the lipid phase.

Lipid rafts [cholesterol- and saturated-fatty-acid-rich membrane patches] are less fluid than surrounding membrane and can cluster scaffold, enzyme, and signaling proteins. *Textbook, Ch. 2, pp. 18-19*

WHY: Keeping interacting signaling proteins near each other can make signaling faster and more organized.

Section Summary Table

Concept	Must know	Exam connection
Differentiation	Selective gene expression + structural specialization	Same genome, different cellular phenotype.
Membrane asymmetry	Inner and outer leaflets differ in lipid composition	Each membrane face has specialized roles.

Concept	Must know	Exam connection
Trilaminar TEM	Dark-light-dark membrane profile after osmium fixation	Hydrophobic center is the pale middle band.
Freeze-fracture	Bilayer splits through hydrophobic core	Reveals embedded membrane proteins.
Lipid rafts	Cholesterol/saturated-lipid-rich signaling patches	Reduced local fluidity + protein clustering.

18. Textbook Expansion - Transport, Endosomal Sorting, Secretion & Cell Signaling

Expanded from the uploaded textbook, Chapter 2, pp. 19-27. This section adds mechanisms and fates that are only partly shown in the slides.

Transport mechanisms

The textbook separates passive membrane transport into **simple diffusion**, **channel-mediated transport**, **carrier-mediated facilitated diffusion**, and **osmosis**. All move material down an appropriate gradient without direct cellular energy expenditure. *Textbook, Ch. 2, pp. 19-24*

WHY: The membrane can allow downhill movement in several ways depending on whether the substance can cross lipid directly or needs a protein pathway.

Channels are multipass membrane proteins that form selective transmembrane pores; many channels open or close in response to physiological stimuli. **Aquaporins** are specialized water channels. *Textbook, Ch. 2, pp. 19-20*

WHY: A pore provides a hydrophilic path through the hydrophobic membrane while still controlling what passes.

Carriers bind selected small molecules and change conformation to move them across the membrane. **Pumps** are active-transport carriers that use energy from ATP hydrolysis to move solutes against gradients. *Textbook, Ch. 2, p. 20*

WHY: Carriers work like revolving doors, whereas pumps pay energy to move cargo uphill.

Endosomal sorting after receptor-mediated endocytosis

In clathrin-mediated endocytosis, receptor-ligand complexes recruit **adaptor proteins** and clathrin; **dynamin** constricts the neck of the forming pit so the coated vesicle can pinch free. *Textbook, Ch. 2, p. 26*

WHY: Clathrin bends the membrane and dynamin helps complete vesicle separation.

Internalized vesicles shed and recycle their clathrin coat, then join the **endosomal compartment**. From there, receptor-ligand complexes can be sent to late endosomes/lysosomes, receptors can be recycled to the surface, or cargo can undergo **transcytosis** [vesicular transport across a cell]. *Textbook, Ch. 2, p. 26*

WHY: The endosome is a sorting station rather than simply a one-way path to lysosomal destruction.

The textbook describes **caveolae** [small flask-shaped membrane invaginations] as another form of receptor-associated uptake, involving **caveolins** and **cavins**. *Textbook, Ch. 2, p. 22*

WHY: Cells can internalize material through more than one specialized membrane domain, not only clathrin-coated pits.

Constitutive vs regulated secretion

Constitutive secretion continuously releases newly synthesized products soon after they are produced, while **regulated secretion** stores products in secretory granules until an appropriate signal triggers exocytosis. *Textbook, Ch. 2, pp. 23-24*

WHY: Some products are needed continuously; others must be released only at the right moment.

The textbook also notes **exosomes** [small extracellular vesicles released by cells], some of which can transfer membrane and contents to other cells as a form of communication. *Textbook, Ch. 2, p. 23*

WHY: Cells can communicate not only with soluble molecules but also by sending packaged molecular cargo.

Signal-receptor classes and amplification

Hydrophilic ligands can signal through three major membrane-receptor classes described in the textbook: **channel-linked receptors**, **enzymatic receptors**, and **G-protein-coupled receptors (GPCRs)**. *Textbook, Ch. 2, pp. 25-27*

WHY: The same basic problem - getting an outside message into the cell - can be solved by opening a channel, activating an enzyme, or activating a G protein.

With **GPCRs**, ligand binding activates a G protein by promoting GTP binding; the activated G protein then regulates an effector such as an ion channel or **adenylyl cyclase**, which can generate cAMP. *Textbook, Ch. 2, pp. 25-27*

WHY: The receptor does not need to enter the cell; it changes an internal G protein that relays and amplifies the message.

The textbook distinguishes **autocrine** signaling [a cell signals itself] and **juxtacrine** signaling [a membrane-bound signal acts on a directly contacting cell], in addition to longer-range signaling patterns. *Textbook, Ch. 2, p. 25*

WHY: Signal range determines whether a message acts on the same cell, an adjacent cell, or a more distant target.

Steroid and thyroid hormones are lipophilic, travel with carrier proteins in plasma, cross cell membranes, and bind intracellular receptors; steroid hormone-receptor complexes can enter the nucleus and alter gene transcription.

Textbook, Ch. 2, p. 26

WHY: Lipid-soluble signals can bypass surface receptors and directly influence gene expression through intracellular receptor proteins.

Section Summary Table

Concept	Must know	Exam connection
Channel / carrier / pump	Pore / conformational transporter / ATP-driven transporter	Know passive vs active logic.
Endosome	Sorting compartment after endocytosis	Recycle receptor, degrade cargo, or transcytose.
Dynamin	Pinches off coated vesicles	Works with clathrin pathway.
Constitutive vs regulated	Immediate continuous release vs stored signal-triggered release	Secretory-granule cells use regulated secretion.
GPCR	GTP-binding G protein activates effectors	Can generate cAMP and amplify signaling.
Caveolae	Caveolin/cavin-rich invaginations	Alternative specialized uptake domain.

19. Textbook Expansion - Ribosomes, RER Protein Quality Control & Smooth ER Specialization

Expanded from the uploaded textbook, Chapter 2, pp. 27-31. Adds important protein-processing details beyond the slide outline.

Ribosomes and cytoplasmic basophilia

A functional ribosome uses mRNA as the template and tRNAs carrying amino acids to assemble a polypeptide. Multiple ribosomes translating one mRNA form a **polyribosome / polysome**. *Textbook, Ch. 2, p. 27*

WHY: mRNA supplies the order and tRNA supplies the amino acids; many ribosomes on one message accelerate output.

An abundance of free or RER-bound polyribosomes can produce **basophilic cytoplasm** [cytoplasm staining blue-purple with basic dyes] because rRNA is strongly basophilic. *Textbook, Ch. 2, p. 27*

WHY: Cells rich in protein-synthesis machinery contain large amounts of RNA, which binds basic stains.

RER targeting, translocation, folding and quality control

Proteins entering the RER pathway begin with a **signal peptide** recognized by **signal recognition particle (SRP)**; SRP pauses/redirects translation to an SRP receptor and translocon in the RER membrane. *Textbook, Ch. 2, pp. 28-29*

WHY: The signal peptide is an address label that routes the growing protein to the ER instead of leaving it in the cytosol.

As the polypeptide enters the RER cisterna, its signal sequence may be removed and the protein undergoes folding and early posttranslational modification. *Textbook, Ch. 2, pp. 28-29*

WHY: A newly synthesized protein is not automatically functional; it must be shaped and chemically processed.

RER **molecular chaperones** [proteins that assist correct folding] and enzymes monitor protein folding; improperly folded proteins can be retained and directed toward degradation rather than exported. *Textbook, Ch. 2, pp. 28-29*

WHY: Quality control prevents defective proteins from entering the secretory pathway.

The textbook emphasizes that RER is especially prominent in cells actively secreting proteins and that proteins for secretion, membranes, and some organelles enter this pathway. *Textbook, Ch. 2, pp. 28-30*

WHY: A cell making large amounts of exportable protein needs a large protein-folding and processing factory.

SER specialization

SER is prominent in cells specializing in lipid synthesis, steroid-hormone production, detoxification, glycogen metabolism, or Ca²⁺ storage. *Textbook, Ch. 2, pp. 30-31*

WHY: The same organelle is expanded differently depending on the chemical work a cell performs.

Detoxification reactions in SER can chemically modify lipid-soluble drugs and toxins so they become easier to eliminate; repeated exposure can increase SER abundance and detoxifying capacity. *Textbook, Ch. 2, pp. 30-31*

WHY: Cells can adapt to chemical workload by expanding the organelle that performs detoxification.

In muscle, specialized SER called the **sarcoplasmic reticulum** [calcium-storing smooth ER of muscle] sequesters and rapidly releases Ca²⁺ during contraction. *Textbook, Ch. 2, p. 31*

WHY: Muscle contraction needs a rapid reusable calcium reservoir located close to contractile machinery.

Section Summary Table

Concept	Must know	Exam connection
Polysome	Many ribosomes on one mRNA	High output of one protein message.
SRP/signal peptide	Targets growing protein to RER	Entry into secretory/endomembrane pathway.
Chaperones	Assist and monitor protein folding	Quality control before export.
SER detox	Modifies lipid-soluble compounds	SER can expand with increased detox demand.
Sarcoplasmic reticulum	Specialized muscle SER for Ca ²⁺	Rapid contraction control.

20. Textbook Expansion - Golgi Processing, Secretory Granules, Lysosomes & Proteasomal Quality Control

Expanded from the uploaded textbook, Chapter 2, pp. 31-38. Adds processing, sorting and degradation detail beyond the slides.

Golgi maturation and trafficking

Cargo arriving from RER enters the **cis Golgi**, moves through medial Golgi cisternae, and leaves through the **trans Golgi network** after sequential enzymatic processing. *Textbook, Ch. 2, pp. 31-33*

WHY: Golgi polarity turns the organelle into an assembly line with different processing enzymes at different stages.

Golgi modifications include trimming or adding carbohydrate groups, **glycosylation**, and **sulfation**; these modifications can alter protein stability, function, recognition, and destination. *Textbook, Ch. 2, pp. 31-33*

WHY: Chemical processing can change both what a protein does and where the cell sends it.

Transport vesicles use coat proteins that help determine direction and destination; the textbook summary specifically emphasizes **COPII** and **COPI** during Golgi-related trafficking. *Textbook, Ch. 2, p. 51*

WHY: Vesicle coats function like routing machinery that helps cargo move between the correct compartments.

Secretory granules [dense Golgi-derived storage vesicles] concentrate regulated secretory products and remain in the cytoplasm until a signal causes exocytosis. *Textbook, Ch. 2, pp. 33-34*

WHY: This lets a cell separate production from release and deliver a large pulse of product quickly.

Lysosomal digestion and recycling

Primary lysosomes leave the Golgi containing acid hydrolases; after fusion with endocytic or phagocytic vesicles, the resulting **secondary lysosomes** become heterogeneous digestive compartments. *Textbook, Ch. 2, pp. 34-36*

WHY: The enzyme package and the material-to-be-digested are made separately and become active together after fusion.

Lysosomal digestion releases useful small molecules back to the cytoplasm for reuse, while indigestible material can remain in **residual bodies**. *Textbook, Ch. 2, pp. 34-36*

WHY: The lysosome is both a waste-disposal unit and a recycling center.

During **autophagy**, membrane surrounds an old or unnecessary organelle to form an autophagosome, which then fuses with a lysosome for degradation. *Textbook, Ch. 2, pp. 35-37*

WHY: The cell can recycle its own worn-out components instead of waiting for them to fail.

Proteasomal protein quality control

Proteasomes degrade many short-lived, damaged, misfolded, or unnecessary proteins in the cytoplasm and nucleus after the proteins are tagged with **ubiquitin**. *Textbook, Ch. 2, p. 37*

WHY: Ubiquitin creates a selective label so the degradation machinery destroys chosen proteins rather than random proteins.

Proteasomal degradation is ATP-dependent and differs from lysosomal digestion: proteasomes are non-membranous protein complexes, whereas lysosomes are acidic membrane-bound organelles that digest material delivered in vesicles. *Textbook, Ch. 2, p. 37*

WHY: The two systems both remove unwanted material but use different substrates, compartments, and mechanisms.

Section Summary Table

Concept	Must know	Exam connection
Golgi polarity	Cis -> medial -> trans	Processing is sequential.
Secretory granule	Stored concentrated regulated product	Rapid signal-triggered secretion.
Secondary lysosome	Primary lysosome fused with cargo vesicle	Active digestion compartment.
Autophagy	Lysosomal recycling of the cell's own	Internal renewal/quality control.

Concept	Must know	Exam connection
	organelles	
Proteasome	Ubiquitin-directed protein degradation	Non-membranous; ATP-dependent.

21. Textbook Expansion - Mitochondrial Bioenergetics & Peroxisomal Metabolism

Expanded from the uploaded textbook, Chapter 2, pp. 38-42. Adds core energy and oxidation mechanisms that are only abbreviated in the slides.

Mitochondrial compartments and ATP production

Mitochondria are abundant in cells and subcellular regions with high energy demand, and they grow/divide by **fission** of preexisting mitochondria. *Textbook, Ch. 2, pp. 38-39*

WHY: Energy-hungry cells maintain more ATP-producing machinery, and mitochondria reproduce from existing organelles rather than being made from scratch.

The mitochondrial **matrix** contains enzymes for fatty-acid beta-oxidation and the **citric acid (Krebs) cycle** [a central fuel-oxidation pathway], which generate reduced electron carriers such as NADH and FADH₂. *Textbook, Ch. 2, pp. 38-39*

WHY: The matrix strips high-energy electrons from fuels and loads them onto carriers that deliver energy to the inner membrane.

The inner mitochondrial membrane contains the **electron-transport system** and **ATP synthase**. Electron transfer is coupled to proton movement, creating an electrochemical gradient used by ATP synthase to make ATP. *Textbook, Ch. 2, pp. 38-39*

WHY: The electron-transport chain builds a stored proton gradient, and ATP synthase converts that gradient into chemical energy in ATP.

The outer mitochondrial membrane is relatively porous, whereas the inner membrane is highly selective; this difference is essential for maintaining the proton gradient across the inner membrane. *Textbook, Ch. 2, pp. 38-39*

WHY: ATP synthesis would fail if protons could freely leak back across the inner membrane.

The textbook links mitochondria to apoptosis: stressed mitochondria can release **cytochrome c**, triggering a regulated cascade that culminates in cell death. *Textbook, Ch. 2, pp. 38-39*

WHY: Mitochondria are not only power plants; they also participate in deciding when a severely damaged cell should die.

Peroxisomal oxidative chemistry

Peroxisomal oxidases perform oxidation reactions that generate H₂O₂, and **catalase** converts H₂O₂ to less harmful products. *Textbook, Ch. 2, pp. 39-41*

WHY: The organelle contains both the reaction that makes peroxide and the enzyme that neutralizes it, keeping reactive chemistry contained.

Peroxisomes participate in oxidation of fatty acids and other metabolic pathways and can form by growth/division of existing peroxisomes as well as from precursor membranes related to ER. *Textbook, Ch. 2, pp. 39-41*

WHY: Peroxisomes are dynamic organelles that can expand their population when oxidative workload increases.

Most peroxisomal proteins are synthesized on free cytosolic ribosomes and then imported using targeting signals recognized by peroxisomal receptors. *Textbook, Ch. 2, pp. 39-41*

WHY: Like mitochondria, peroxisomes import many completed proteins from the cytosol instead of using RER-bound ribosomes.

Section Summary Table

Concept	Must know	Exam connection
Matrix	Krebs cycle + beta-oxidation enzymes	Generates NADH/FADH ₂ .
Inner membrane	ETC + ATP synthase	Builds/uses proton gradient to make ATP.
Cytochrome c	Can trigger apoptosis when released	Links mitochondrial stress to cell death.

Concept	Must know	Exam connection
Peroxisomal oxidase	Generates H ₂ O ₂	Oxidative metabolism.
Catalase	Breaks down H ₂ O ₂	Limits oxidative damage.

22. Textbook Expansion - Microtubule Dynamics, Centrosomes & Intracellular Transport

Expanded from the uploaded textbook, Chapter 2, pp. 42-45. Adds dynamic-instability and polarity concepts that are high-yield but minimally developed in the slides.

Microtubule polarity and dynamic instability

Microtubules are polarized polymers of alpha/beta-tubulin heterodimers with a **plus end** that generally grows more rapidly and a **minus end** commonly anchored near a microtubule-organizing center. *Textbook, Ch. 2, pp. 42-44*

WHY: Polarity gives the microtubule a built-in direction, which is crucial for directed motor-protein traffic.

Microtubules show **dynamic instability** [rapid switching between growth and shrinkage] as GTP-tubulin is added and lost. *Textbook, Ch. 2, pp. 43-44*

WHY: A cytoskeletal track that can quickly grow or collapse can reorganize the cell for transport, movement, and mitosis.

Microtubule-associated proteins (MAPs) can stabilize microtubules and help organize them into functional arrays.

Textbook, Ch. 2, pp. 43-44

WHY: Tubulin alone forms the tube, while associated proteins tune how stable and organized that tube is.

Motor proteins and centrosome

Microtubule polarity determines direction of vesicle and organelle transport by **kinesin** and **dynein**; the textbook assessment explicitly tests polarity as the basis for transport direction. *Textbook, Ch. 2, pp. 43-45, assessment p. 52*

WHY: The motors recognize the oriented track, so the plus/minus ends function like directional signs.

The **centrosome** [major microtubule-organizing center near the nucleus] contains a pair of centrioles at right angles in a matrix rich in tubulin and other proteins. Before mitosis the centrosome duplicates, and the two centrosomes become spindle poles. *Textbook, Ch. 2, p. 45*

WHY: The centrosome provides an organized origin for the radial microtubule network and later helps build the mitotic spindle.

Each centriole contains nine microtubule triplets, matching the slide emphasis, but the textbook additionally places centrioles within the functional centrosome that organizes spindle microtubules. *Textbook, Ch. 2, p. 45*

WHY: The triplet cylinder is the structural component; the surrounding centrosomal material performs much of the actual microtubule nucleation.

Section Summary Table

Concept	Must know	Exam connection
Polarity	Plus end vs minus end	Defines directional transport.
Dynamic instability	Alternating growth and shrinkage	Allows rapid cytoskeletal remodeling.
MAPs	Stabilize/organize microtubules	Tune microtubule behavior.
Centrosome	Main MTOC with paired centrioles	Duplicates before mitosis; spindle poles.
Kinesin/dynein	Opposite-direction motors	Use microtubule polarity for traffic.

23. Textbook Expansion - Actin, Myosins & Intermediate-Filament Mechanics

Expanded from the uploaded textbook, Chapter 2, pp. 44-48. Adds cytoskeletal mechanisms and protein classes not fully developed on the slides.

Actin filament dynamics and functions

Actin filaments are thin, polarized polymers of G-actin subunits assembled into F-actin. Continuous addition at one end and removal at the other can produce **treadmilling** [subunit flow through a filament of roughly constant length].

Textbook, Ch. 2, pp. 44-46

WHY: The filament can remain similar in length while constantly renewing its building blocks, allowing rapid remodeling.

Actin-binding proteins regulate polymerization, disassembly, severing, capping, cross-linking, membrane attachment, and bundling. Examples shown in the textbook include profilin, cofilin, gelsolin, filamin, Arp2/3, spectrin, villin, fimbrin, and alpha-actinin. *Textbook, Ch. 2, pp. 45-46*

WHY: Actin's function depends heavily on helper proteins that decide whether filaments form networks, bundles, short fragments, or membrane supports.

The textbook links actin remodeling to changes in cytoplasmic viscosity, endocytosis, cell cleavage after mitosis, and cell locomotion. *Textbook, Ch. 2, pp. 45-46*

WHY: Actin is the cytoskeleton's rapid shape-changing system, so it is used wherever the cell must deform or move.

Myosins [actin-based motor proteins] bind and move along actin filaments, carrying vesicles or generating movement; myosin I is associated with membranes, while myosin II is important in contractile structures. *Textbook, Ch. 2, pp. 45-46*

WHY: Actin provides the track and myosin converts ATP energy into force along that track.

Intermediate-filament classes and mechanical strength

Intermediate filaments are the most stable major cytoskeletal polymers and primarily provide mechanical strength rather than acting as tracks for motor proteins. *Textbook, Ch. 2, pp. 46-48*

WHY: Stable rope-like filaments help cells resist stretching and tearing.

The textbook groups intermediate-filament proteins by cell type: **cytokeratins** in epithelial cells, **desmin** in muscle, **GFAP** in astrocytes, **vimentin** in mesenchymal cells, **neurofilaments** in neurons, and **lamins** in nuclei. *Textbook, Ch. 2, p. 47*

WHY: Because different tissues express characteristic intermediate filaments, these proteins can also identify cell lineage and specialized mechanical needs.

Keratin filaments provide strong mechanical support to epithelial cells; the textbook assessment uses epidermolysis bullosa simplex to illustrate that keratin defects cause tissue fragility. *Textbook, Ch. 2, pp. 47-52*

WHY: If the main reinforcement cables of an epithelial cell are defective, minor mechanical stress can damage the tissue.

Section Summary Table

Concept	Must know	Exam connection
Actin treadmilling	Subunits add at one end and leave at the other	Dynamic remodeling without major length change.
Actin-binding proteins	Control assembly, branching, bundling, capping	Determine filament architecture.
Myosin	Actin-based motor	Force generation and short-range cargo movement.
Intermediate filaments	Most stable cytoskeletal polymers	Mechanical strength, not vesicle tracks.
Keratin disease link	Keratin defects -> epithelial fragility	Mechanical-stability principle.

24. Textbook Expansion - Cellular Inclusions & Histologic Interpretation

Expanded from the uploaded textbook, Chapter 2, pp. 48-50. Adds the textbook's distinctions among storage products and pigments.

Inclusions are not organelles

Cellular inclusions are cytoplasmic deposits or stored materials that are not metabolically active organelles and are not present in all cells. *Textbook, Ch. 2, pp. 48-50*

WHY: They are better thought of as stored supplies, pigments, or leftover material than as active cellular machines.

Lipid droplets are hydrophobic lipid stores surrounded by a phospholipid **monolayer** rather than a conventional bilayer; associated proteins and enzymes regulate lipid metabolism. *Textbook, Ch. 2, p. 48*

WHY: A hydrophobic fat core does not need a full watery compartment, so one phospholipid layer is sufficient to separate it from cytosol.

Glycogen granules [stored glucose polymer] are important readily available energy reserves and can form characteristic aggregates in cells with high metabolic activity. *Textbook, Ch. 2, p. 48*

WHY: Glycogen packages many glucose units into a compact store that can be mobilized when energy demand rises.

Lipofuscin [yellow-brown “wear-and-tear” pigment] represents accumulating residual material from lysosomal digestion and is especially notable in long-lived cells. *Textbook, Ch. 2, p. 48*

WHY: A cell that survives for decades has more time to accumulate indigestible leftovers.

Melanin [dark pigment that absorbs light] can help protect nuclei from DNA damage caused by light, while **hemosiderin** [iron-containing storage pigment] can accumulate in cells involved in iron handling. *Textbook, Ch. 2, pp. 48-50*

WHY: Different pigments reflect different functions or histories: photoprotection for melanin and iron storage/turnover for hemosiderin.

The textbook summary table also contrasts surface specializations: **microvilli** increase absorptive surface area and are supported by actin microfilaments, whereas **cilia** and the sperm **flagellum** are membrane extensions supported by microtubules. *Textbook, Ch. 2, pp. 49-50*

WHY: Surface shape reflects function: actin-supported microvilli maximize area, while microtubule-based cilia/flagella are built for movement.

Section Summary Table

Concept	Must know	Exam connection
Lipid droplet	Neutral lipid core + phospholipid monolayer	Energy/lipid storage.
Glycogen	Stored glucose polymer granules	Rapid energy reserve.
Lipofuscin	Indigestible lysosomal residual pigment	Accumulates in long-lived cells.
Melanin	Light-absorbing pigment	Protects against light-related DNA damage.
Hemosiderin	Iron-containing pigment	Reflects iron storage/turnover.
Microvilli vs cilia	Actin-based absorption vs microtubule-based movement	Do not confuse surface specializations.

25. Slide-by-Slide Coverage Map

This map confirms that every slide in both uploaded slide decks was accounted for in the notes.

Cell 1 deck - 27 slides

Cell 1 - Slide 1: Title slide - CELL; establishes topic.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 1 - Slide 2: Specialized cellular functions table - Section 1.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 1 - Slide 3: Cell = cytoplasm + nucleus; cytosol/organelles - Section 1.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 1 - Slide 4: Plasma membrane composition/thickness - Section 2.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 1 - Slide 5: Membrane diagram and four membrane functions - Section 2.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 1 - Slide 6: Cholesterol, bilayer asymmetry, carbohydrate receptor functions - Section 2.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 1 - Slide 7: Integral and peripheral proteins - Section 2.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 1 - Slide 8: Glycocalyx - Section 2.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 1 - Slide 9: Passive/active membrane transport - Section 3.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 1 - Slide 10: Endocytosis overview - Section 3.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 1 - Slide 11: Phagocytosis - Section 3.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 1 - Slide 12: Pinocytosis - Section 3.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 1 - Slide 13: Receptor-mediated endocytosis/clathrin - Section 3.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 1 - Slide 14: LDL receptor uptake and receptor recycling diagram - Section 3.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 1 - Slide 15: Exocytosis, Ca²⁺, membrane trafficking - Section 3.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 1 - Slide 16: Hydrophilic signals and second messengers - Section 4.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 1 - Slide 17: Hydrophobic signals and intracellular receptors - Section 4.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 1 - Slide 18: Nucleus overview - Section 5.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 1 - Slide 19: Nucleus diagram, components, regulation and ribosomal production - Section 5.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 1 - Slide 20: Nuclear envelope and nuclear pores micrographs - Section 5.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 1 - Slide 21: Nuclear lamina and functions - Section 5.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 1 - Slide 22: Nucleolus zones and rRNA/ribosome functions - Section 5.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 1 - Slide 23: Chromatin composition; DNA/histone/nonhistone material - Section 5.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 1 - Slide 24: Heterochromatin vs euchromatin and percentages - Section 5.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 1 - Slide 25: MCQ: nucleolus is site of rRNA transcription - Section 5.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 1 - Slide 26: MCQ: hydrophobic membrane center forms water-soluble barrier - Section 2.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 1 - Slide 27: Resource slide - source list acknowledged here; no new study facts.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 deck - 42 slides

Cell 2 - Slide 1: Title slide - CYTOPLASMIC ORGANELLES.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 2: Cytoplasm composition/pH/sol-gel transformation - Section 6.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 3: Cytoplasmic inclusions - Section 6.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 4: Membranous vs non-membranous organelles - Section 6.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 5: Ribosome structure/polysomes/5'→3' movement - Section 7.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 6: Free vs ER-bound polyribosome targeting and ubiquitin-proteasome branch - Section 7.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 7: RER structure and destinations - Section 8.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 8: SRP-mediated movement of polypeptides into RER - Section 8.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 9: RER histology visual reinforcement - Section 8.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 10: SER structure and lipid synthesis - Section 8.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 11: SER functions - Section 8.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 12: Jaundice and osteogenesis imperfecta applications - Section 8.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 13: Golgi structure/faces/modifications - Section 9.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 14: COP-II forward; COP-I retrograde + Golgi EM - Section 9.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 15: Detailed RER-to-Golgi processing/sorting diagram - Section 9.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 16: I-cell disease and M6P tagging defect - Section 9.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 17: Lysosome basics/pH/primary-secondary lysosomes - Section 10.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 18: Lysosomal storage disease table - Section 10.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 19: Residual bodies diagram/histology - Section 10.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 20: Peroxisomes and deficiency disorders - Section 11.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 21: Proteasomes and protein aggregates - Section 11.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 22: Mitochondrial outer membrane/intermembrane compartment - Section 12.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 23: Inner membrane/cristae/matrix - Section 12.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 24: Plate-like vs tubular cristae - Section 12.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 25: Maternal mtDNA, granules, MERRF - Section 12.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 26: Cytoskeleton introduction visual - Section 13.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 27: Microtubule roles/structure - Section 13.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 28: Kinesin vs dynein - Section 13.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 29: MTOC and regulation - Section 13.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 30: Chediak-Higashi syndrome - Section 13.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 31: Centriole nine-triplet organization - Section 13.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 32: Centriole/centrosome diagram - Section 13.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 33: Microfilament/actin images - Section 14.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 34: G-actin/F-actin and dynamics - Section 14.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 35: Intermediate-filament image/role - Section 14.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 36: Vimentin/desmin/GFAP/neurofilament/keratin - Section 14.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 37: Intrinsic apoptosis text - Section 15.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 38: Intrinsic, extrinsic, and cytotoxic-T-cell apoptosis diagram - Section 15.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 39: Apoptotic morphology and lack of inflammation - Section 15.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 40: MCQ: HPV E6/p53 -> prevention of apoptosis - Section 15.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 41: MCQ: highest mitochondrial content -> cardiac muscle - Section 12.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Cell 2 - Slide 42: Resource slide - source list acknowledged here; no new study facts.

WHY: This entry confirms the slide was reviewed and incorporated; title/resource or visual-only slides are identified rather than treated as new factual content.

Source boundary

All factual note content above is drawn from the uploaded slide decks and uploaded textbook PDF. No web or outside reference material was added.

WHY: This keeps the study set aligned to the material you were actually given and minimizes conflicts with your course wording.

The only material intentionally allowed to go beyond direct document wording is the simple **WHY** explanation, as requested; those explanations are designed to make the source facts intuitive rather than introduce new examinable facts.

WHY: The WHY is a teaching aid, not a second source of course content.

Resource slides included in the decks

Both decks cite Mescher & Junqueira's Basic Histology, Lippincott's Illustrated Q&A Review of Histology, and First Aid for the USMLE Step 1; each deck also includes a YouTube link.

WHY: These are the resources listed by the slide author; they are recorded here for completeness but were not independently searched or used as external sources.

[End of Notes]